

DSA:

UNIT 1:

Structure declaration, initialisation and accessing with example – 12m

Structure declaration using pointers with example – 8m

Dynamic memory allocation(all three functions) with example – 12m

Self - referential structures with example – 8m

Classification of data Structures(Types and syntax) – 8m

Abstract Data Type – 8m

Time and space complexity(Very important) – 12m

UNIT 2:

Array implementation of list and their operations: Traversing, Insertion and Deletion(algorithm and syntax) (either of two operations – 8m, all operations – 12m)

Linked list's implementation and their operations: Insertion, Deletion and Search(Algorithm and syntax) – 12m

Double linked list and their operations(Insertion and Deletion) – 8m

Josephus Problem – 8m

Polynomial and Arithmetic operations on it, write algorithm and example program(very important) – 12m

Sparse Matrix – 8m

UNIT 3:

Stack and its operations with example – 8m

Stack implementation using array with example – 8m

Stack implementation using linked list with example – 8m

Infix to postfix conversion (Very important) – 12m

Postfix evaluation and balancing symbols – 4m or 8m

Recursion and its example – 8m

Tower of Hanoi (Very important) – 8m

Queue implementation using linked list – 8m

Priority Queue with example – 8m

UNIT 4:

Binary Tree traversal (three types, inorder, preorder and postorder) – 12m

Binary Search Trees and its operations – 12m

AVL tree – 8m

Hash functions – 8m

Collision Handling – 8m

UNIT 5:

BFS – 8m

DFS – 8m

Prim's algorithm – 8m

Kruskal's algorithm – 8m

Dijkstra's algorithm(diagram, cost and steps) (Very important) – 12m